

NSW Electricity Market · Internal Briefing

What Drives the Daily Price Curve

Half-Hourly price, generation and storage dynamics across a representative week

Keiren Brandt-Sawdy July 2025

Source: OpenElectricity (AEMO data) · NSW1 · 11–18 July 2025

NSW prices swing ~\$227/MWh a day:

The renewable and storage mix sets the trend

\$227
/MWh

swing of the average
daily price

—
-\$22 to \$300/MWh weekly range

\$93/MWh weekly average

1 Solar collapses midday prices

Abundant rooftop and utility solar pushes prices in the middle of the day to ~\$4/MWh - occasionally negative

2 Storage mitigates the swing

Batteries and pumped hydro charge cheaply in the middle of the day and discharge into the evening peak, providing a firming function like gas

3 Cheaper wind pushes out expensive gas

On windy days, wind generation pushes the price down and less gas enters the system

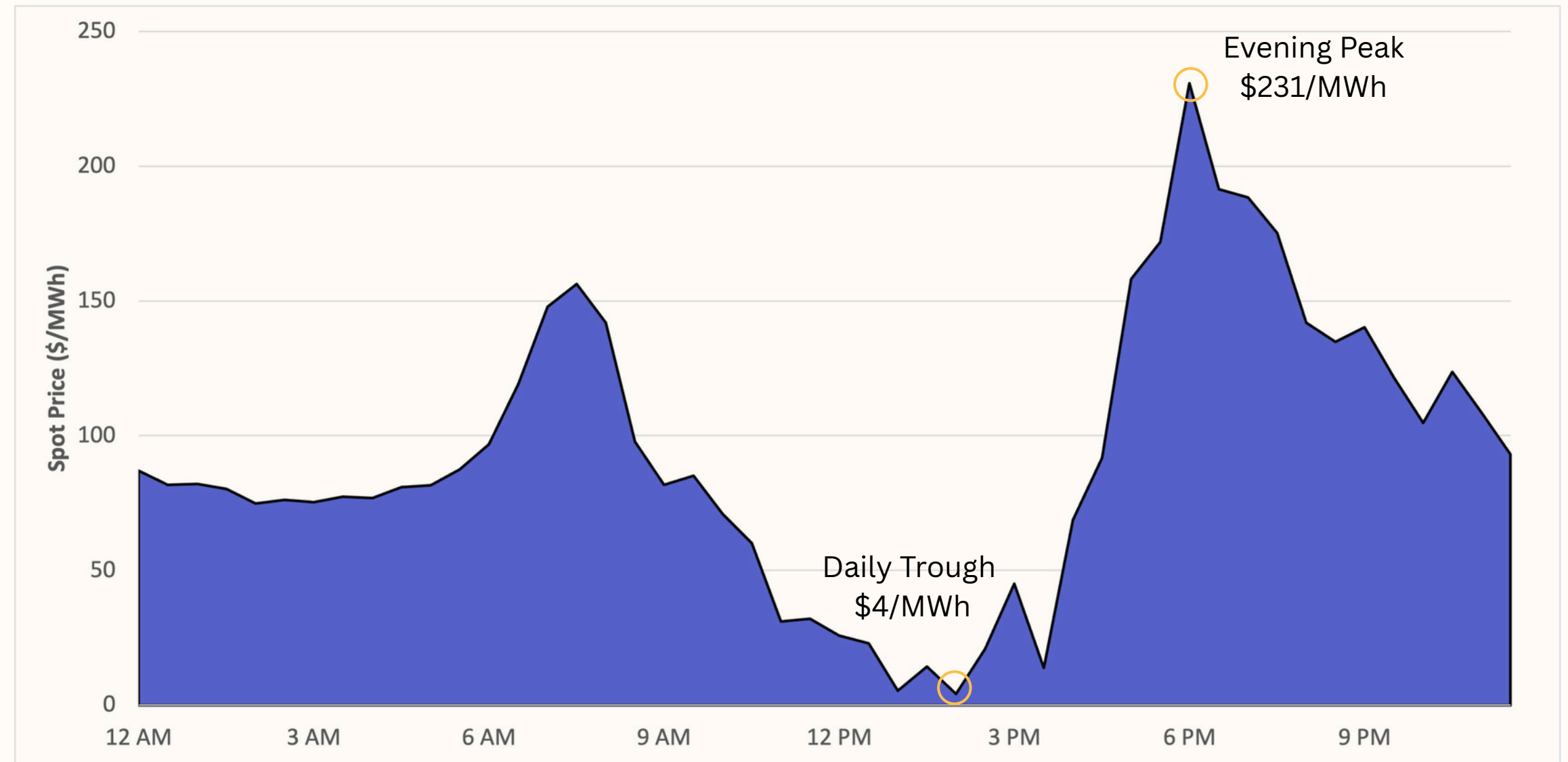
Prices crater in the middle of the day and spike in the evening

\$4

Average daytime trough

\$231

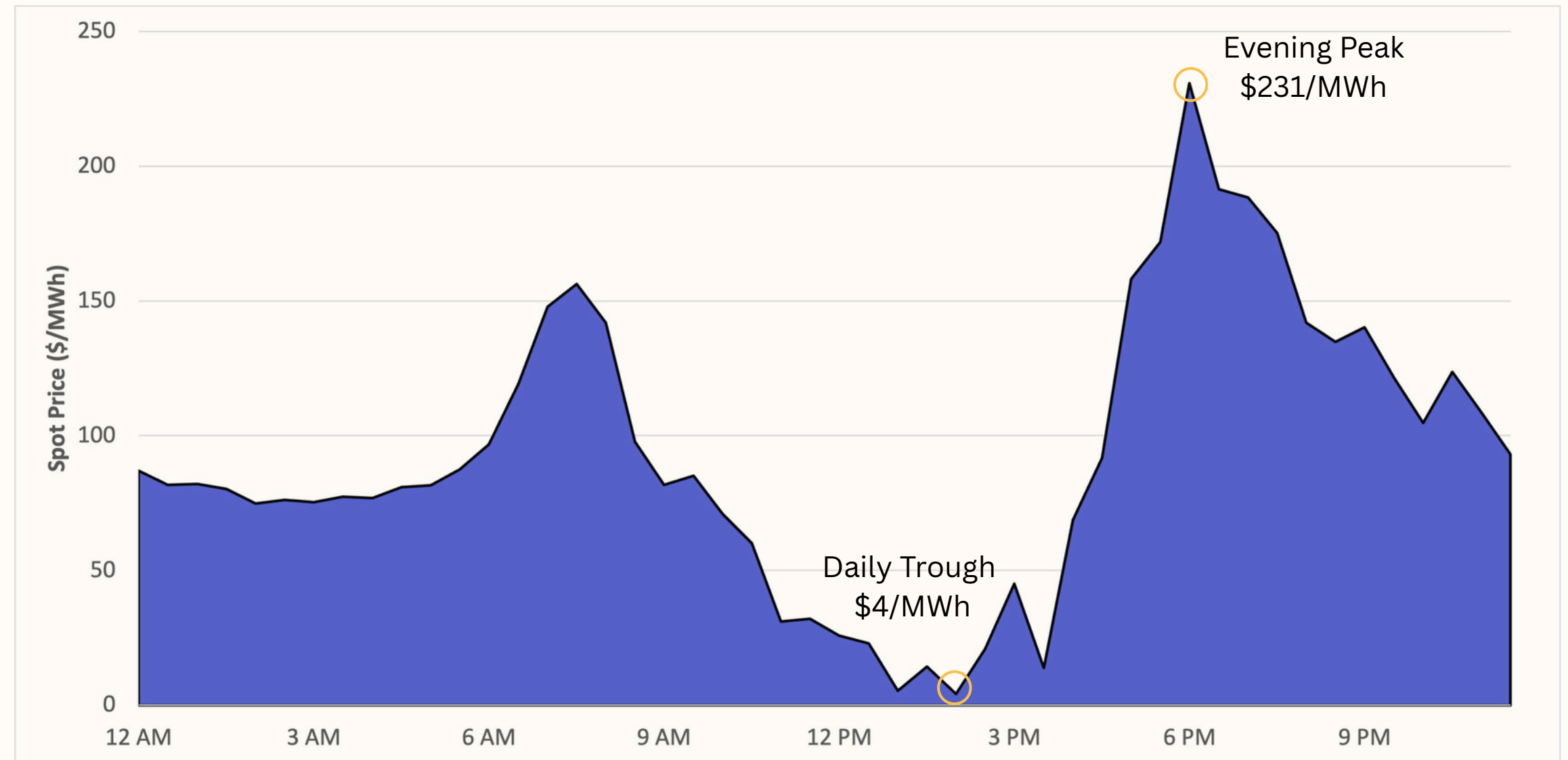
Average evening peak



Prices crater in the middle of the day and spike in the evening

THIS MEANS...

Electricity prices have a predictable daily shape, the classic “duck curve”, that repeats across the week.



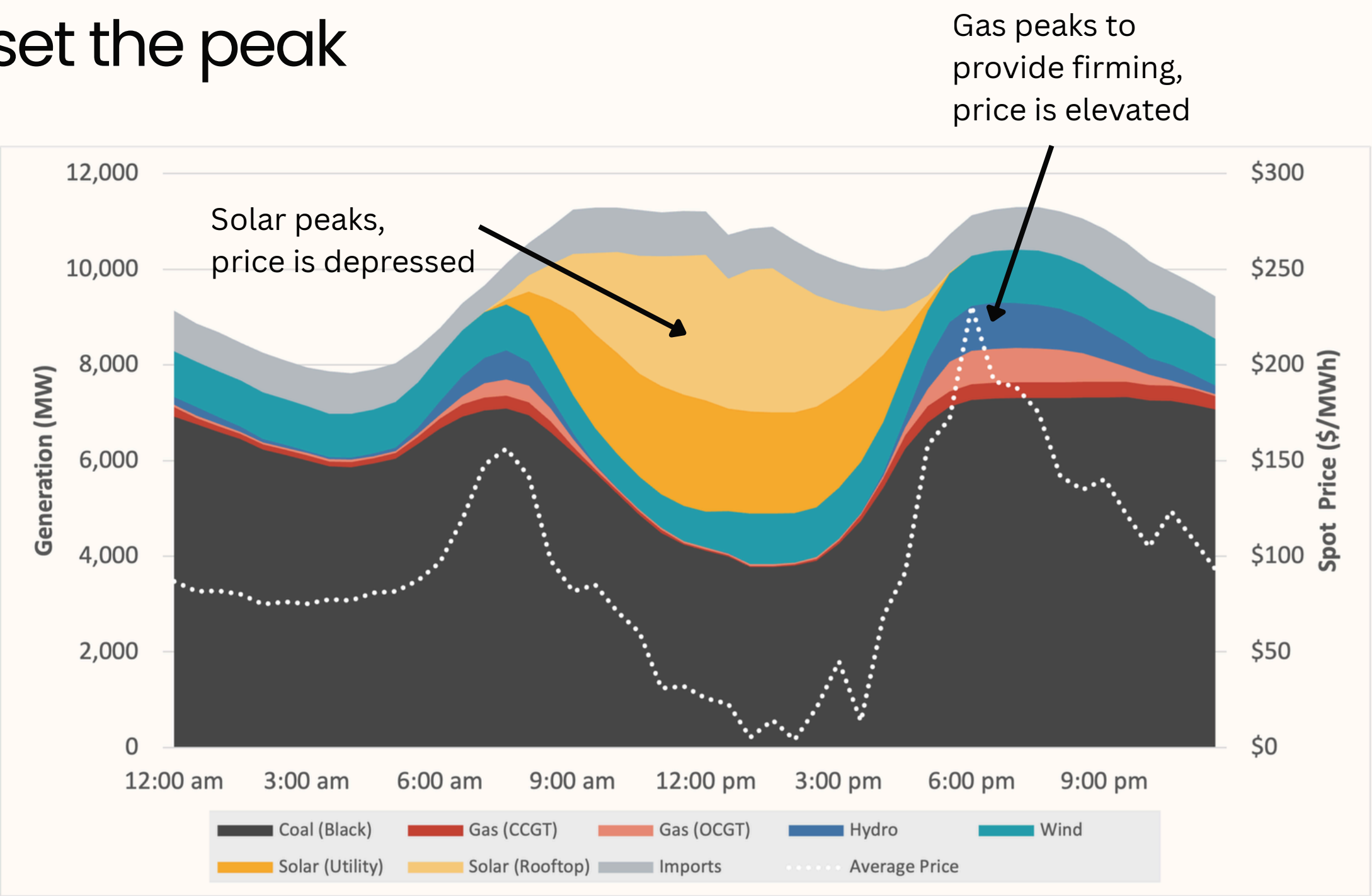
Cheap midday solar pushes prices down
its exit in the evening lets gas set the peak

5,376 MW

Solar peak at Midday

38 → 718 MW

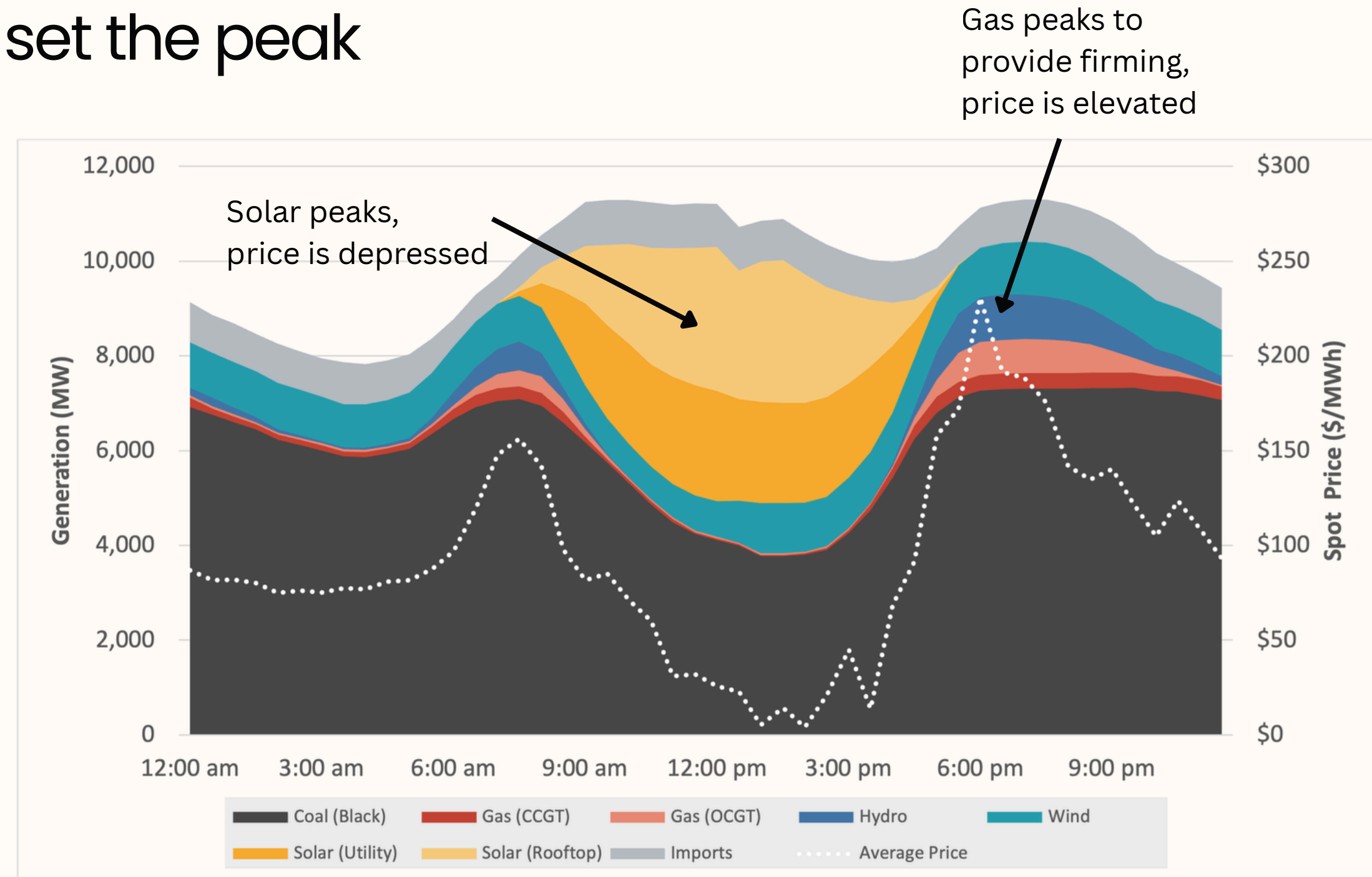
Gas (OCGT) evening ramp



Cheap midday solar pushes prices down its exit in the evening lets gas set the peak

THIS MEANS...

As solar exits, coal cannot ramp fast enough. Gas bids in and sets the higher marginal price.

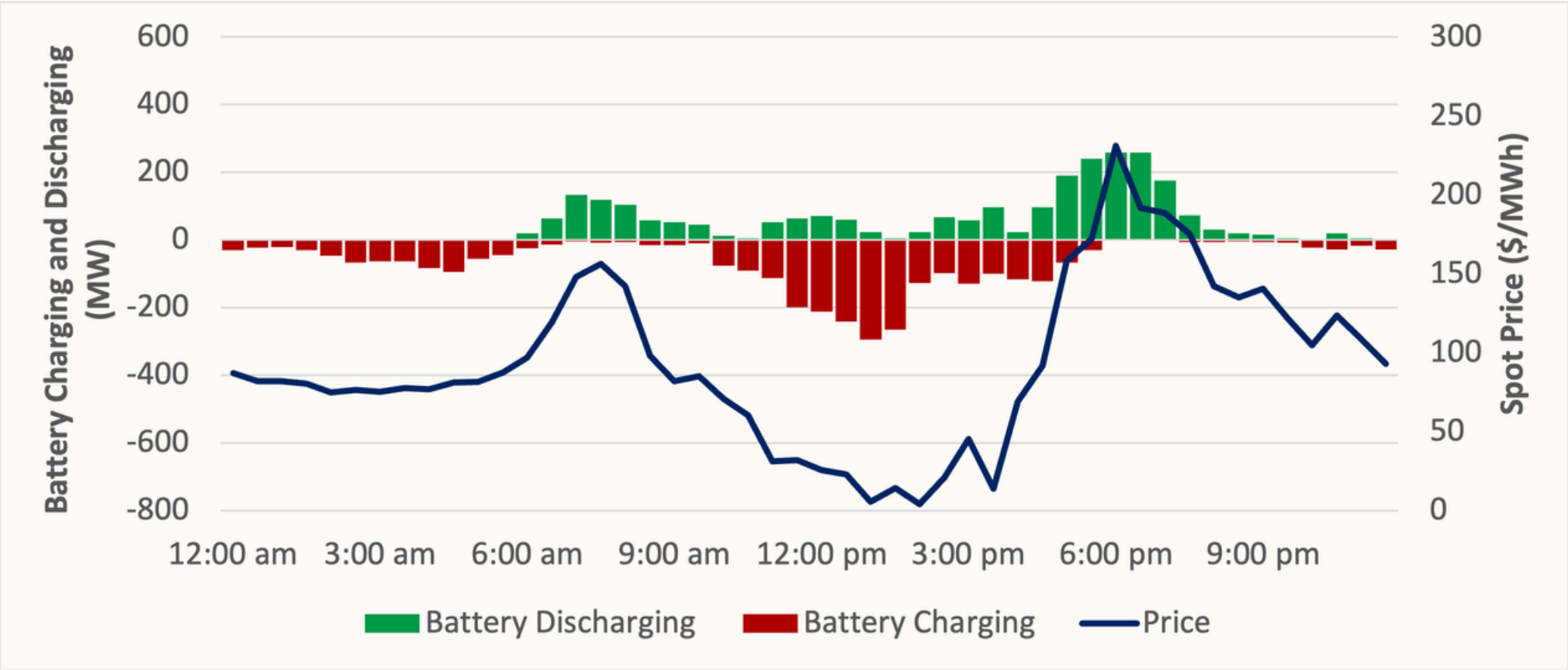


Finding 3 – Dispatchable Renewables

Batteries and Hydro act as a renewable “peaker”

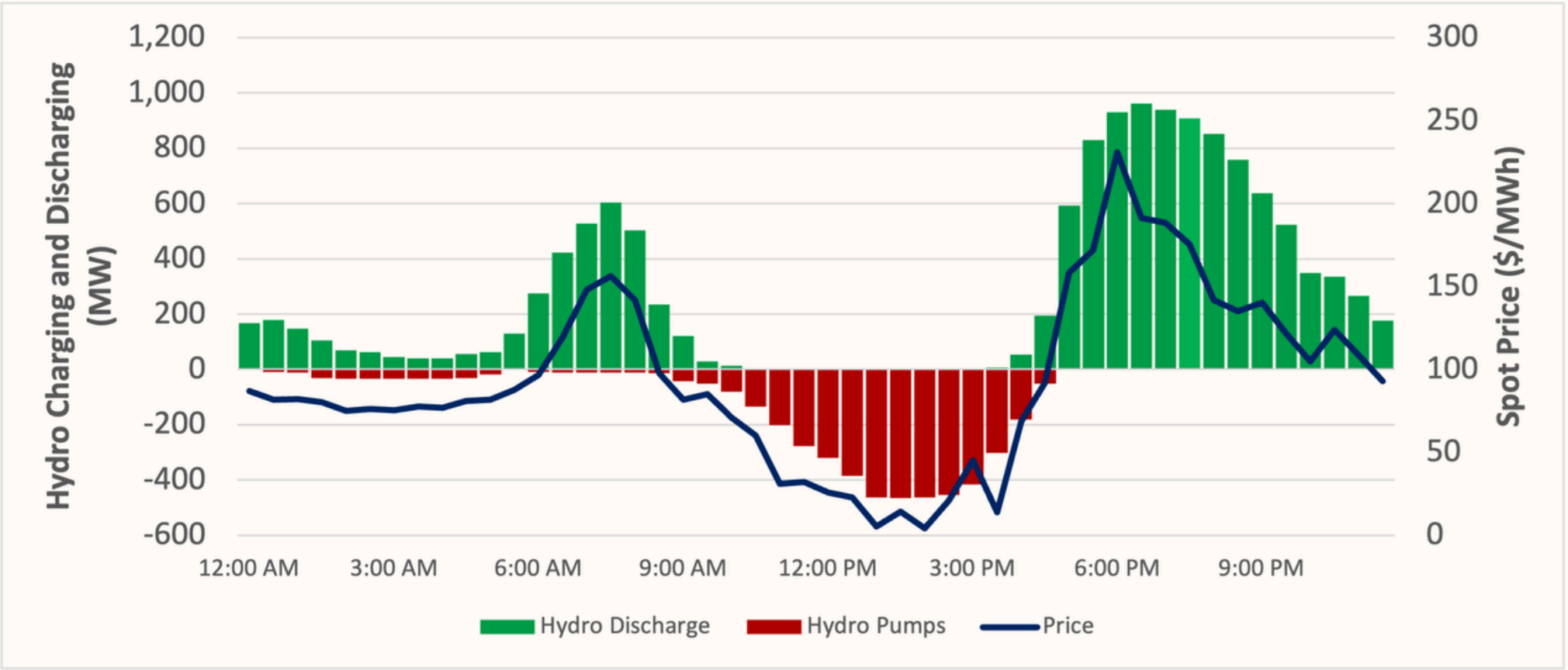
-294 → 258 MW

Batteries day charge to evening ramp



-466 → 962 MW

Hydro day charge to evening ramp

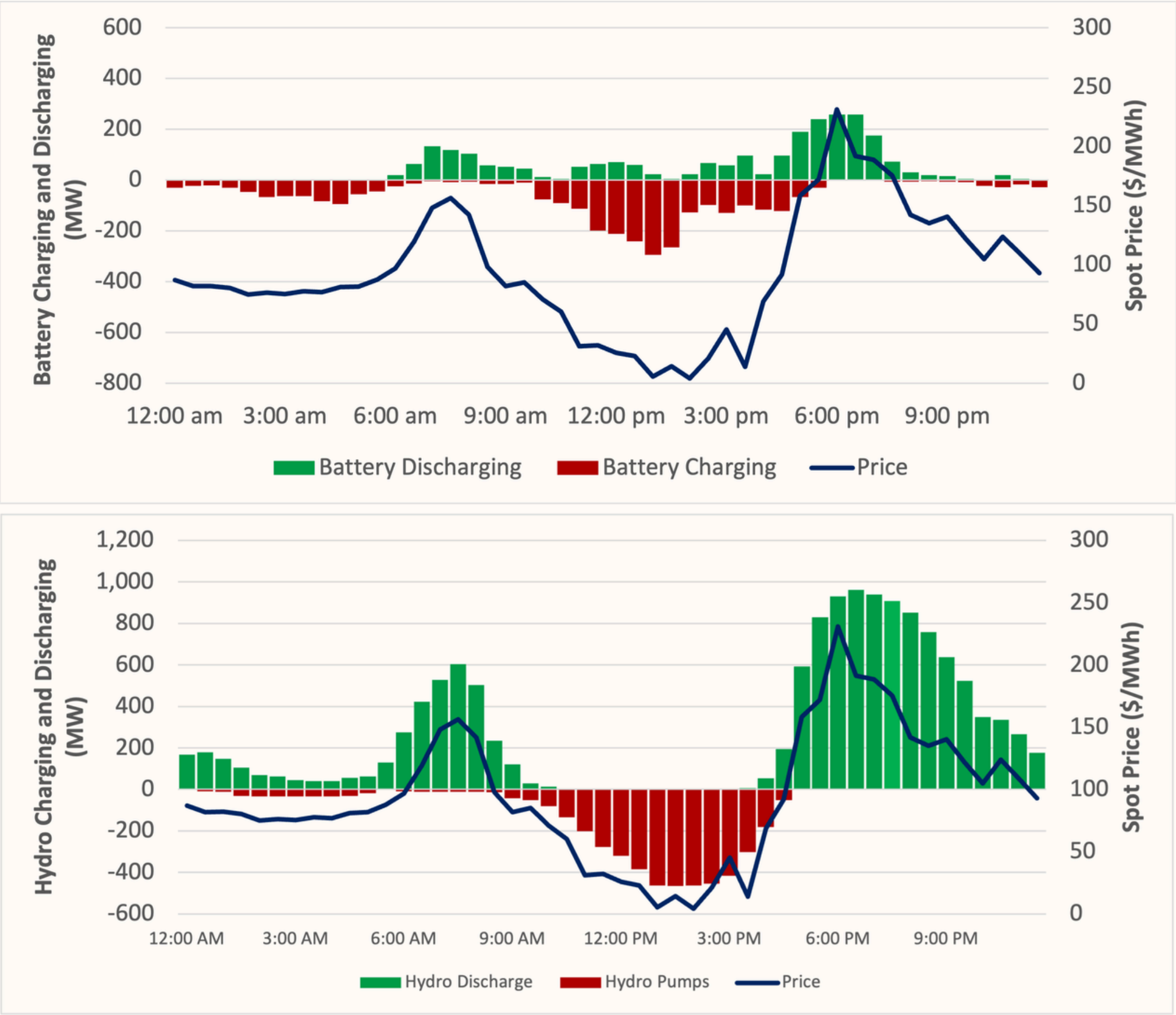


Finding 3 – Dispatchable Renewables

Batteries and Hydro act as a renewable “peaker”

THIS MEANS...

Batteries and hydro charge in the middle of the day when prices are cheap, and discharge into the evening peak. They provide a firming capacity similar to gas



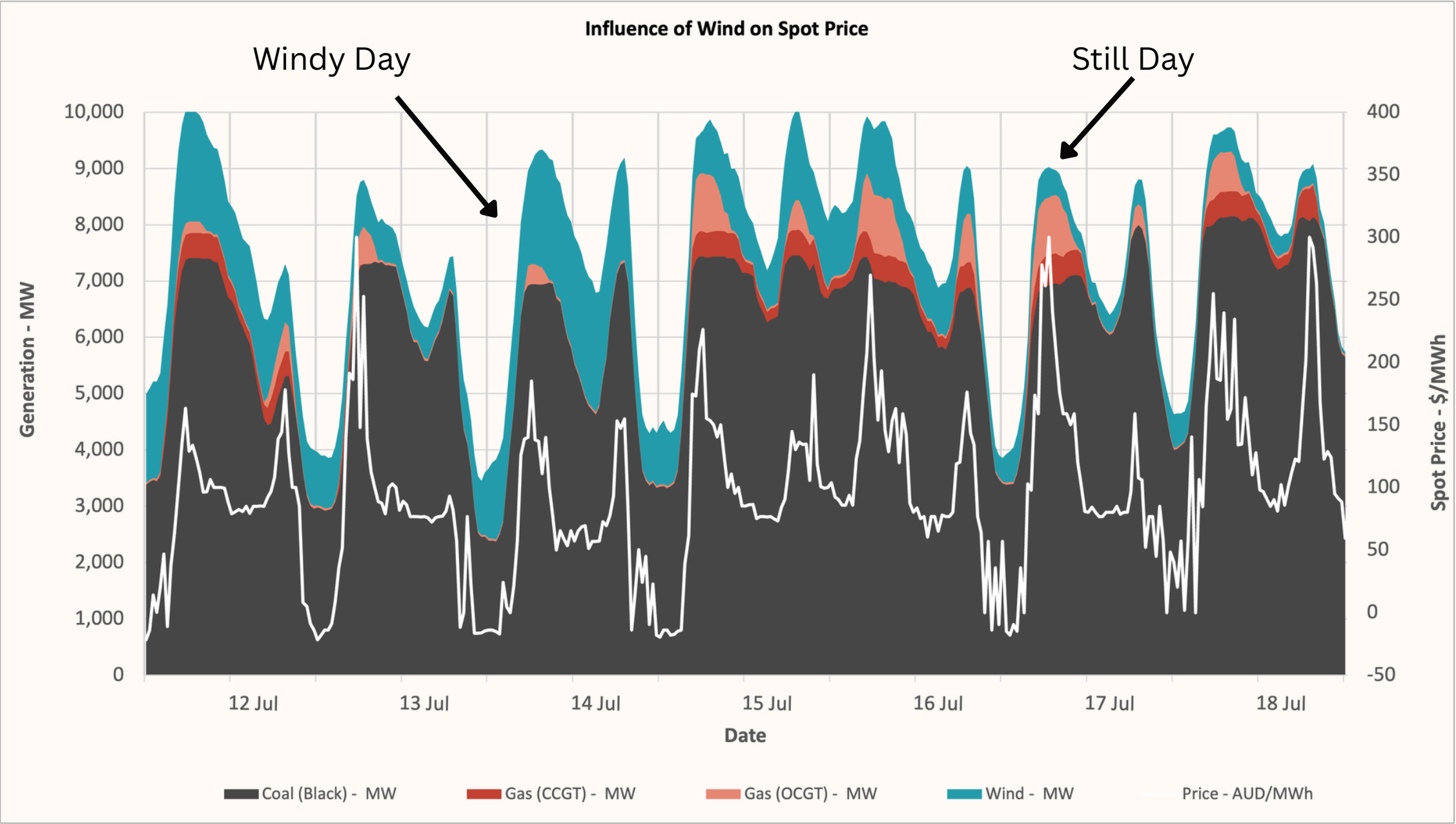
More evening wind means lower peak prices

\$185

Peak on a windy day (July 13 at 6.30pm)

\$300

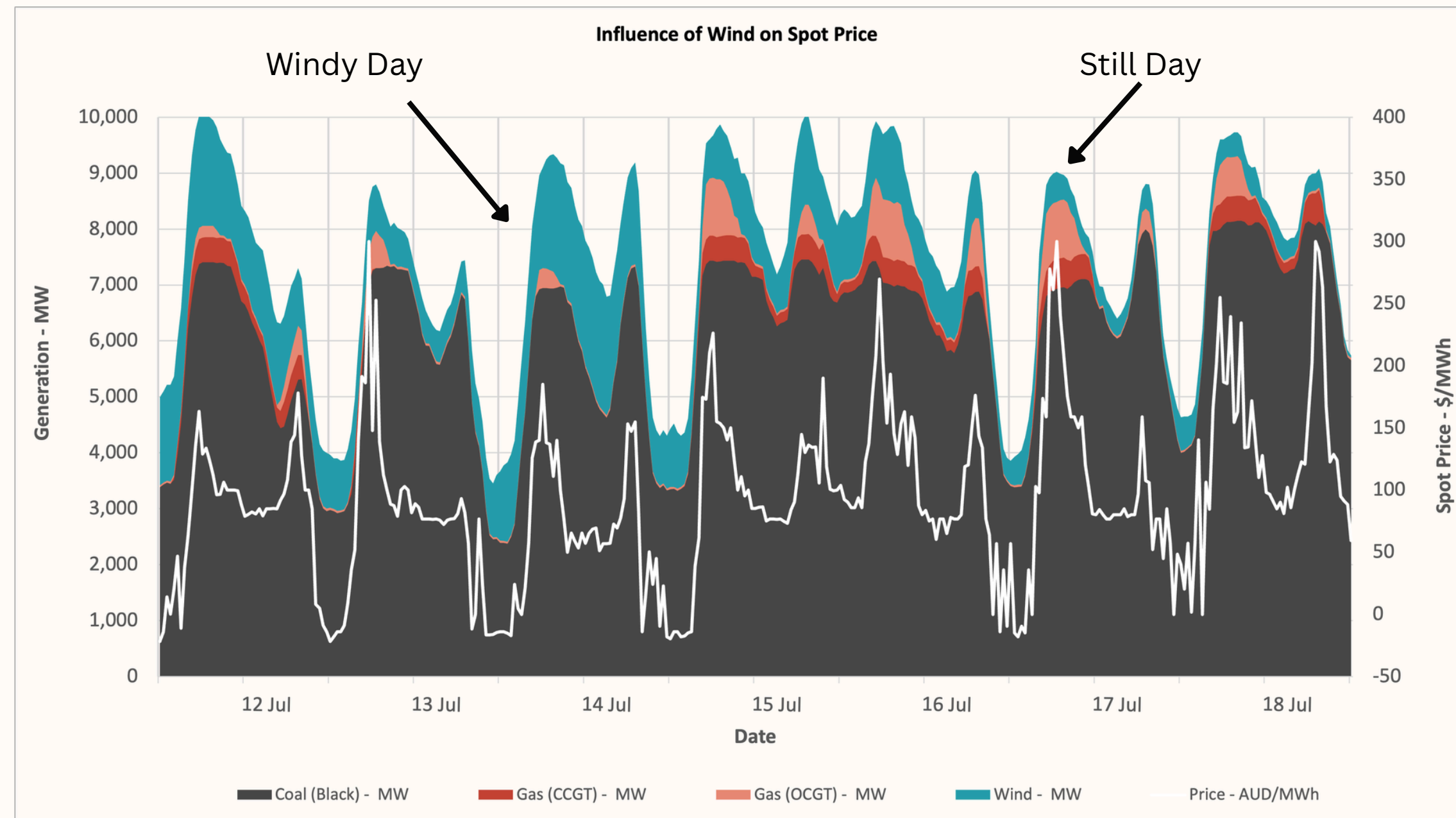
Peak on a still day (July 16 at 7.00pm)



More evening wind means lower peak prices

THIS MEANS...

Wind is variable, but when it blows into the evening it displaces expensive gas and caps the peak.



What this means...

The midday-to-evening spread is a structural, repeatable feature of the market. That makes it something to plan around.

Move demand to daytime when solar depresses prices

Shifting hot water, EV charging and pool pumps to midday cuts bills and absorbs surplus solar

Firm the evening peak when gas spikes prices

Gas sets the evening price, wind moderates that spike. That window is the system's tightest and most expensive.

Build storage to capitalise on the price spread

The \$227/MWh gap between trough and peak is the revenue case for batteries and pumped hydro. They charge when power is cheap in the day, and discharge in the evening helping to moderate prices

Appendix

Data and Method

Source	OpenElectricity
Region	New South Wales
Period	11–18 July 2025 (one winter week)
Granularity	30-minute intervals, 337 observations
Metrics	Generation by fuel type (MW), spot price (AUD/MWh), emissions (tCO ₂ e), temperature (°C)
Method	Intraday profiles are averages across the week; the windy vs still comparison uses complete days (13 vs 16 July)
Caveat	A single winter week. Seasonal demand and generation patterns differ across the year